

EMANUEL MICHELE SODA

Milan, Italy · +39 340 875 1226 · emanuelsoda@gmail.com · github.com/EmanuelSoda · linkedin.com/in/emanuel-soda

SUMMARY

Computational scientist with 5+ years of experience developing machine learning, Bayesian statistical models, and reproducible data pipelines for multimodal biomedical data, including spatial transcriptomics, single-cell genomics, digital pathology, live cell imaging, and clinical outcome modelling.

SKILLS

Programming & Data	Git/GitHub, Docker, Python, Pandas, NumPy, Scanpy, R, Tidyverse, SQL, Seurat
Machine & Deep Learning	Scikit-learn, statsmodels, PyTorch & PyTorch Lightning, Pyro, Tidymodels, scvi-tools
Computing & Infrastructure	Bash/Linux, HPC, Nextflow, Weights & Biases
Image & Spatial Analysis	Segmentation, Registration, Spatial Clustering, Squidpy, SpatialData, LazySlide, OpenCV
Pathway Analysis	KEGG, Reactome

EXPERIENCE

Visiting PhD Student April 2026 – Present

Wellcome Sanger Institute (Hinxton, UK)

- Designed multimodal deep learning frameworks integrating GeoMx RNA profiles and H&E; histology features to characterise crypt-level heterogeneity across IBD and healthy controls.

Doctoral Researcher (PhD in Data Analytics and Decision Sciences) September 2023 – Present

Human Technopole & Politecnico di Milano (Milan, Italy)

- Developed reproducible computational pipelines for multimodal IBD patient data integrating spatial transcriptomics, histopathology, cell type composition, and clinical outcome variables.
- Characterised spatially resolved tissue microenvironments across 180 IBD and healthy control samples using probabilistic cell type deconvolution and NMF-based niche discovery.
- Built Bayesian classification and regression models linking spatial cellular composition to patient-level clinical outcomes, supporting interpretable biomarker discovery from Visium data.
- Developed computational pathology workflows integrating instance segmentation, supervised tissue classification, and self-supervised representation learning to quantify crypt-level morphological heterogeneity.
- Applied mixed-effects differential expression modelling to treatment-naïve IBD tissue, identifying dysregulated transcriptional programs across disease subtypes.

Bioinformatician & ML Scientist October 2020 – September 2023

Istituto Europeo di Oncologia (IEO) (Milan, Italy)

- Developed and validated an automated ML pipeline for cell cycle phase classification from live cell imaging data, resulting in a co-authored eLife publication.
- Analysed bulk and single-cell Acute Myeloid Leukaemia data linking metabolic alterations to epigenetic alterations.
- Reduced manual processing time by 80% by replacing visual annotation and manual image handling with automated computational analysis.

EDUCATION

Visiting PhD student (Bayraktar group) April 2026 – Present

Wellcome Sanger Institute, Hinxton, UK

PhD in Data Analytics and Decision Sciences

September 2023 – Present

Human Technopole & Politecnico di Milano, Milan, Italy

MSc in Bioinformatics for Computational Genomics

October 2019 – December 2021

Politecnico di Milano & University of Milan, Milan, Italy

- Developed ScreenR, an R/Bioconductor package for automated hit identification in shRNA loss-of-function screens.

BSc in Computer Science

October 2016 – October 2019

University of Milan Bicocca, Milan, Italy

PUBLICATIONS

Sara Ometto, Soumick Chatterjee, ..., **Emanuel M. Soda**, ..., Craig A. Glastonbury. Hundreds of cardiac MRI traits derived using 3D diffusion autoencoders share a common genetic architecture. *Nature Communications*, 2026.

Kourosh Hayatigolkhatmi*, Chiara Soriani*, **Emanuel M. Soda*** et al. Automated workflow for the cell cycle analysis of (non-)adherent cells using a machine learning approach. *eLife*, 2024.

CONFERENCES & PRESENTATIONS

ECDP 2025 — Oral presentation

Barcelona, June 2025 — Charting the molecular and Spatial Transcriptomic landscape of Crypt variability in IBD.

PhD Symposium "Research in Motion"

Milan, September 2025 — Poster: Crypt variability in IBD.

ASHG 2024 — Poster

Denver, November 2024 — Charting the molecular and Spatial Transcriptomic landscape of Crypt variability in IBD.

AWARDS & GRANTS

First Position — Time-Series Exploration with Deep Learning Hackathon

2024

Milan, September 2024

Best Project Award — Bioinformatics Hackathon 2021 (Protein Edition)

2021

Copenhagen, July 2021

TEACHING & MENTORING

Master's thesis supervision (x2)

2025–2026

Bioinformatics for Computational Genomics, Milan

Introduction to Computational Biology (seminar)

2020–2021

Molecular pharmacology & immunology, University of Milan

SUMMER SCHOOLS

Computer Vision for Automated Medical Diagnosis

September 2025

ICCV 2025, Honolulu

MLX Fundamentals & MLX Health and Bio

July 2024

University of Oxford, UK